

EX-14 Write the Python program to Implement the Alpha-Beta Pruning Algorithm for Gaming

AIM

To develop a Python program to implement the **Alpha-Beta Pruning Algorithm** for gaming, which optimizes the Minimax algorithm by eliminating branches that do not affect the final decision.

ALGORITHM

Step 1: Start the program.

Step 2: Read the terminal node values of the game tree.

Step 3: Initialize **Alpha** = $-\infty$ and **Beta** = $+\infty$.

Step 4: Begin evaluation from the root node.

Step 5: If the current node is a maximizing node, update the Alpha value with the maximum score obtained.

Step 6: If the current node is a minimizing node, update the Beta value with the minimum score obtained.

Step 7: If **Alpha** $\geq$ **Beta**, prune (ignore) the remaining branches.

Step 8: Continue the process until all required nodes are evaluated.

Step 9: Display the optimal value obtained after Alpha-Beta pruning.

Step 10: Stop the program.

PROGRAM CODE

```
import math
```

```
def alphabeta(depth, nodeIndex, maximizingPlayer, values, alpha, beta, maxDepth):
```

```
    if depth == maxDepth:
```

```
        return values[nodeIndex]
```

```
if maximizingPlayer:
    best = -math.inf

    for i in range(2):
        val = alphabeta(depth + 1,
                        nodeIndex * 2 + i,
                        False,
                        values,
                        alpha,
                        beta,
                        maxDepth)

        best = max(best, val)
        alpha = max(alpha, best)

        if beta <= alpha:
            break

    return best

else:
    best = math.inf
```

```
for i in range(2):
    val = alphabeta(depth + 1,
                    nodeIndex * 2 + i,
                    True,
                    values,
                    alpha,
                    beta,
                    maxDepth)
```

```
    best = min(best, val)
```

```
    beta = min(beta, best)
```

```
    if beta <= alpha:
```

```
        break
```

```
return best
```

```
values = list(map(int, input("Enter 8 terminal values: ").split()))
```

```
height = int(math.log2(len(values)))
```

```
result = alphabeta(0, 0, True, values,
                  -math.inf, math.inf, height)
```

```
print("\nOptimal Value :", result)
```

OUTPUT

```
=====
Enter 8 terminal values: 3 5 2 9 12 5 23 23
```

```
Optimal Value : 12
```

RESULT

The Python program successfully implements the **Alpha-Beta Pruning Algorithm** and finds the optimal value while reducing the number of nodes evaluated by pruning unnecessary branches..